Robotic revision surgery after failed Nissen anti-reflux surgery: a single center experience and a literature review.

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The gastroesophageal reflux disease (GERD) worldwide prevalence is increasing maybe due to population aging and the obesity epidemic. Nissen fundoplication is the most common surgical procedure for GERD with a failure rate of approximately 20% which might require a redo surgery. The aim of this study was to evaluate the short- and long-term outcomes of robotic redo procedures after anti-reflux surgery failure including a narrative review. We reviewed our 15-year experience from 2005 to 2020 including 317 procedures, 306 for primary, and 11 for revisional surgery. Patients included in the redo series underwent primary Nissen fundoplication with a mean age of 57.6 years (range, 43-71). All procedures were minimally invasive and no conversion to open surgery was registered. The meshes were used in five (45.45%) patients. The mean operative time was 147 min (range, 110-225) and the mean hospital stay was 3.2 days (range, 2-7). At a mean follow-up of 78 months (range, 18-192), one patient suffered for persistent dysphagia and one for delayed gastric emptying. We had two (18.19%) Clavien-Dindo grade IIIa complications, consisting of postoperative pneumothoraxes treated with chest drainage. Redo anti-reflux surgery is indicated in selected patients and the robotic approach is safe when it is performed in specialized centers, considering its surgical technical difficulty.

Laparoscopic Nissen fundoplication: a five-year single-center clinical experience and results.

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The current surgical approach in the treatment of hiatal hernia with gastroesophageal reflux disease is known as hernioplasty together with antireflux surgical procedures. Among the antireflux surgical treatment procedures, the most applied approach is the laparoscopic Nissen fundoplication. In this study, we aimed to examine the results and effectiveness of laparoscopic Nissen fundoplication and to share our clinical experiences. Patients who underwent laparoscopic Nissen fundoplication operation between January 2017 and January 2022 in the general surgery clinic of a tertiary healthcare center were included in the study. The clinical data, preoperative, operative, and postoperative findings and results of the cases were investigated. The mean age of the patients was 46.2 ± 14.7 years, and the female/male ratio was 1.5/1. According to the Clavien-Dindo classification system, 9.9% of the patients had grade I, and 18.3% grade II complications. The patients were followed up for a mean of 32.6 ± 14.8 months. During the follow-up, reoperation was planned in 5.6% of the patients due to recurrence. Laparoscopic Nissen fundoplication is a well-defined technique. It is a safe and effective surgical method with appropriate patient selection.

Effectiveness of Nissen fundoplication versus anterior and posterior partial fundoplications for treatment of gastro-esophageal reflux disease: a systematic review protocol.

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The objective of this review is to determine the relative effectiveness of Nissen fundoplication compared to anterior and posterior partial fundoplication in controlling the symptoms of gastro-esophageal reflux disease and reducing their side effect profile in adults.The specific questions posed by this review are: what is the effectiveness of Nissen fundoplication in comparison to anterior partial fundoplication (90 degree, 120 degree and 180 degree) and posterior 270 degree fundoplication in terms of symptom control of gastro-esophageal reflux disease, and what are the side effects of these surgical interventions?

Is there a role for anything other than a Nissen's operation?

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The Nissen fundoplication is the most frequently applied antireflux operation worldwide. The aim of this review was to compare laparoscopic Nissen with partial fundoplication. Nine randomized trials comparing several types of wraps were analyzed, four for the comparison Nissen vs. Toupet and five for the comparison Toupet or Nissen vs. anterior fundoplication. Similar comparisons in nonrandomized studies were also included. Dysphagia rates and reflux recurrence were not related to preoperative esophageal persistalsis independent of the selected procedure. Overall, Nissen fundoplication revealed slightly better reflux control, but was associated with more side effects, such as early dysphagia and gas bloat. Advantages of an anterior approach were only reported by one group. A significantly higher reflux recurrence rate for anterior fundoplication was observed in all other comparisons. Tailoring antireflux surgery according to esophageal motility is not indicated. At present, the relevant factor for selection of a Nissen or Toupet fundoplication is personal experience. Anterior fundoplication offers less effective long-term reflux control.

Laparoscopic Nissen fundoplication is an effective treatment for gastroesophageal reflux disease.

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The open Nissen fundoplication is effective therapy for gastroesophageal reflux disease. In this study, the outcomes in 198 patients treated with the laparoscopic Nissen fundoplication was evaluated for up to 32 months after surgery to ascertain whether similar positive results could be obtained. To ensure surgical success, patients were required to have mechanically defective sphincters on manometry and increased esophageal acid exposure on 24-hour pH monitoring. The patients either had severe complications of gastroesophageal reflux disease or had failed medical therapy. These requirements have been found to be necessary to ensure a successful surgical outcome. The disease was complicated by ulceration (46), stricture (25) and Barrett's esophagus (33). Patients underwent standard Nissen fundoplications identical in every detail to open procedures except that the procedures were carried out by the laparoscopic route. Perioperative complications included gastric or esophageal perforation (3), pneumothorax (2), bleeding (2), breakdown of crural repair (2) and periesophageal abscess (1). The only mortality occurred from a duodenal perforation. Six patients required conversion to the open procedure. The median hospital stay was 3 days. One hundred patients were observed for follow-up for 6 to 32 months (median 12 months), with outcomes similar to the open Nissen fundoplication. Further surgery was required for two patients who had recurrent gastroesophageal reflux and one who developed an esophageal stricture. Ninety-seven percent are satisfied with their decision to have the operation. The laparoscopic Nissen fundoplication can be carried out safely and effectively with similar positive results to the open procedure and with all of the advantages of the minimally invasive approach.

Effects of Nissen fundoplication on gastro-oesophageal reflux and oesophageal motor function.

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Nissen fundoplication reduces gastro-oesophageal reflux effectively, but the mechanisms through which this effect is brought about have remained rather obscure. In this study the effect of fundoplication on oesophageal acid exposure, oesophageal body motility, and lower oesophageal sphincter pressure (LOSP) was assessed prospectively. Eleven patients were studied before and 3 months after a floppy Nissen fundoplication. A Dent sleeve was used to measure LOSP, and ambulatory pH and pressure monitoring were used to evaluate oesophageal motor function. Gastro-oesophageal reflux was significantly decreased after fundoplication without an increase in LOSP. The motor function of the oesophageal body was not affected by the antireflux procedure. Nissen fundoplication is an effective antireflux operation, even though the procedure does not increase LOSP, and the motility pattern of the oesophageal body is not affected by the construction of a floppy fundic wrap.

The laparoscopic Nissen fundoplication--a better operation?

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Laparoscopic Nissen fundoplication has become the method of choice in antireflux surgery replacing its open counterpart before the long-term results of controlled clinical trials were available. METHODS AND AIM: Review of the literature to highlight the long-term results of laparoscopic Nissen fundoplication. Long-term symptom relief regarding significant reflux symptoms of heartburn and regurgitation can be achieved by laparoscopic fundoplication in 84% to 97% and patients' overall satisfaction with the result of their laparoscopic fundoplication surgery is high, ranging from 86% to 96%. The long-term results of randomised trials have shown no statistically significant differences in subjective symptomatic outcome between laparoscopic and open Nissen fundoplication. Complaints regarding the scar, incisional hernias and higher incidence of defective wraps were associated with the open approach. At long-term follow-up the laparoscopic Nissen fundoplication has a similar long-term subjective symptomatic outcome as the open procedure but laparoscopic Nissen fundoplication is associated with a significantly lower incidence of incisional hernias and defective fundic wraps at endoscopy, defining laparoscopic Nissen fundoplication as the procedure of choice in surgical management of gastro-oesophageal reflux disease.

Acute gastric volvulus complicated by gastric perforation following laparoscopic Nissen fundoplication; A case report.

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Laparoscopic Nissen Fundoplication is an effective standard surgical procedure for treatment of severe GERD. While it is generally safe and effective, a rare but potentially fatal complication known as acute gastric volvulus can occur following this procedure. A 28-year-old male, ten months post Laparoscopic Nissen Fundoplication presented with a one-day history of severe epigastric pain, abdominal distention, unproductive retching, and difficulty in breathing. Examination revealed tachypnea, subcutaneous emphysema and a tender distended abdomen. Imaging studies showed a left pneumothorax, pneumoperitoneum, and a grossly distended stomach. Emergency exploratory laparotomy confirmed organoaxial gastric volvulus, necrosis of the greater curvature and gastric perforation. Partial gastrectomy and anterior gastropexy were performed. A left thoracostomy tube was placed to drain the left pneumothorax. He recovered fully post-operatively with complete resolution of all symptoms. Acute Gastric volvulus post Laparoscopic Nissen Fundoplication is attributed to adhesions, gastrostomy tubes, and foreign bodies like sutures. Life-threatening complications, such as gastric perforation, can ensue, underscoring the need for swift diagnosis and treatment. Acute gastric volvulus following Laparoscopic Nissen Fundoplication is a rare condition, and is difficult to diagnose. Given the steadily increasing rates of laparoscopic Nissen fundoplications performed in Uganda, maintaining a high index of suspicion is crucial for favorable patient outcomes among patients with this potentially fatal complication.

Revisional One-Anastomosis Gastric Bypass (OAGB) After Intrathoracic Migration of Nissen Fundoplication.

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According to the latest IFSO recommendations, bariatric and metabolic surgery is the recommended treatment for patients with a BMI above 35 kg/m2 (with or without associated pathology), achieving good results in terms of weight loss in the medium to long term, as well as improving a significant percentage of comorbidities in this type of patient (diabetes mellitus, arterial hypertension, dyslipidaemia, gastro-esophageal reflux disease (GERD)...). The incidence of GERD is higher in patients with obesity, with more severe symptoms. Over the years, Nissen fundoplication has been the gold standard treatment for patients with GERD who do not respond to medical treatment. However, in patients with obesity, gastric bypass is a valid option to consider. We present the case of a patient who had previously undergone anti-reflux surgery (laparoscopic Nissen) for GERD, with favorable evolution, who presented intrathoracic migration of the same after 8 years, with new onset of symptoms, and who was offered revision bariatric surgery. The video presents on the performance of OAGB in a patient who had previously undergone antireflux surgery, with intrathoracic Nissen. Performing this technique after a previous Nissen fundoplication (as well as migration of the Nissen) is a somewhat more complex procedure than primary surgery but can be performed safely with careful technique (there are often previous adhesions that hinder mobility and separation of the fundoplication) and provides good symptom control.

Laparoscopic fundoplication for gastroesophageal reflux disease.

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Gastroesophageal reflux disease (GERD) is a condition that develops when the reflux of gastric contents into the esophagus leads to troublesome symptoms and/or complications. Heartburn is the cardinal symptom, often associated with regurgitation. In patients with endoscopy-negative heartburn refractory to proton pump inhibitor (PPI) therapy and when the diagnosis of GERD is in question, direct reflux testing by impedance-pH monitoring is warranted. Laparoscopic fundoplication is the standard surgical treatment for GERD. It is highly effective in curing GERD with a 80% success rate at 20-year follow-up. The Nissen fundoplication, consisting of a total (360°) wrap, is the most commonly performed antireflux operation. To reduce postoperative dysphagia and gas bloating, partial fundoplications are also used, including the posterior (Toupet) fundoplication, and the anterior (Dor) fundoplication. Currently, there is consensus to advise laparoscopic fundoplication in PPI-responsive GERD only for those patients who develop untoward side-effects or complications from PPI therapy. PPI resistance is the real challenge in GERD. There is consensus that carefully selected GERD patients refractory to PPI therapy are eligible for laparoscopic fundoplication, provided that objective evidence of reflux as the cause of ongoing symptoms has been obtained. For this purpose, impedance-pH monitoring is regarded as the diagnostic gold standard.